

A Major Project Report on

VisuoFind

Submitted in partial fulfillment of the requirements for the degree of Bachelor
of Engineering in Software Engineering at Pokhara University

By

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Lamachaur, Kaski, Nepal

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ABSTRACT

VisuoFind is a reverse image search and product recommendation system designed to enhance the online shopping experience through visual input. The system enables users to upload a product image and receive visually similar product suggestions using deep learning and vector similarity search. The core of VisuoFind lies in extracting meaningful visual features from product images using a pretrained ResNet-50 model, which converts images into high-dimensional embeddings. These embeddings are indexed using FAISS (Facebook AI Similarity Search) for fast and efficient similarity matching. The system also aims to integrate text-based similarity using SBERT (Sentence-BERT) to extract embeddings from product descriptions. A combined approach using both image and text embeddings is planned to improve search relevance. The recommendation engine is designed to be content-based, using user preferences, past searches, and favorites to personalize suggestions. It is developed with the goal of making product discovery faster, easier, and more personalized for users by combining advanced AI models with efficient retrieval techniques.

Keywords: Reverse image search, Content-based recommendation, ResNet-50, SBERT, FAISS, Deep learning, Visual similarity, Personalized product search

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

In the current digital age, e-commerce has become an integral part of people's lives, offering unmatched convenience and efficiency for consumers. Users can browse and purchase products from the comfort of their homes, but the ever-expanding product catalogs on these platforms often lead to decision fatigue and reduced satisfaction. In response, recommendation systems have become crucial in e-commerce platforms. Studies indicate that nearly 75% of users rely on recommendation systems to navigate these extensive catalogs effectively. These systems analyze user data, such as past purchases and browsing behavior, to deliver personalized suggestions, enhancing user experience and satisfaction (Ejjami, 2024).

In addition to recommendation systems, visual search has emerged as a transformative feature, enabling intuitive product discovery by allowing users to search using images rather than text-based descriptions. This innovative approach leverages computer vision to interpret and understand images, connecting users with relevant products more efficiently (Dr T R Arunkumar, 2023).

The VisuoFind (Product Search and Recommendation AI) system aims to seamlessly integrate personalized recommendation systems with advanced visual search capabilities. By combining these features, the system seeks to deliver a highly intuitive, efficient, and engaging browsing experience, ensuring that users can quickly find products that align with their preferences and needs.

1.2 PROBLEM STATEMENT

The rapid growth of e-commerce has significantly increased the size of product catalogs, making it difficult for users to find desired products efficiently. Traditional keyword-based search methods often fall short in addressing user intent, especially when product descriptions are vague or missing. This gap results in user frustration, longer browsing times, and reduced satisfaction, which can negatively impact platform engagement and sales (Technologies, 2023). This project aims to address these issues by developing an AI-based Product Search and Recommendation system

that integrates personalized recommendation algorithms and visual search functionality. The goal is to enhance user experience by enabling intuitive product discovery and increase engagement on e-commerce platforms.

1.3 OBJECTIVES

- To develop a system that enables users to search for products using image recognition and receive personalized product recommendations based on visual and user preference data.

1.4 IMPLICATION

The development of an AI-driven Product Search and Recommendation system for e-commerce platforms has the potential to the online shopping experience by addressing key issues such as decision fatigue and inefficient product discovery. By integrating personalized recommendations with visual search capabilities, this project aims to offer users a more intuitive and efficient way to navigate large product catalogs. The insights gained will deepen our understanding of how artificial intelligence, particularly machine learning and computer vision, can be applied to build smarter, more user-centric e-commerce solutions. Beyond improving user satisfaction and boosting sales, this research contributes to the broader use of AI in practical applications, showcasing its impact on optimizing user experiences across various domains.

CHAPTER 2

LITERATURE REVIEW

Several existing systems and platforms have explored the use of visual search and recommendation technologies in e-commerce. While significant progress has been made using algorithms like CNNs, ResNet, and VGG for image recognition and product matching, very few systems provide an integrated experience that combines visual product search, personalized recommendations, and support for diverse product categories in a single platform.

Amazon:

Amazon is a global e-commerce leader that uses user behavior, purchase history, and browsing patterns to generate highly personalized recommendations. However, Amazon's visual search capabilities are limited, primarily available on mobile devices through the camera-based "StyleSnap" feature, which does not offer full functionality across all product types. It focuses more on recommendations than true visual similarity matching. (Amazon, 2011)

Syte:

Syte provides advanced visual search tools that allow users to find products by uploading images. It uses deep learning and computer vision to match visually similar items. However, its services are mostly built for retailers as B2B tools, rather than offering a consumer-facing, full shopping experience with end-to-end visual discovery and personalized suggestions. (Syte, 2015)

Pinterest Lens:

Pinterest Lens is a popular visual discovery tool that enables users to upload or capture photos to explore visually related content. While powerful in terms of visual matching, it is not specifically designed for shopping or product recommendations. The results are based on general image similarity rather than refined product features or personalization. (Pinterest, 2017)

Zalando:

Zalando, a fashion-focused e-commerce site, combines some visual search features with personalized fashion recommendations. However, it only supports clothing items and lacks the flexibility to work across other product categories. It also does not fully integrate image-based search with advanced AI recommendation engines. (Zalando, 2018)

To compare, the table below summarizes the key features of these platforms with our proposed system:

Table 2.1: Comparison Table

Platform	Visual Search Availability	Personalized Recommendations	Product Diversity
Amazon	Basic visual search (limited to mobile)	Provides tailored product suggestions using purchase history and browsing patterns	Wide category coverage
Syte	Advanced AI-based visual matching	Offers personalized recommendations based on user preferences	Primarily limited to a narrow range of products
Pinterest Lens	Excels in visual discovery, helping users find related images	Recommendations aren't limited to products	Too diverse and isn't limited to a certain category
Zalando	Visual search for fashion items	Provides styling suggestions based on profiles	Focused solely on clothing products
VisuoFind (Proposed)	AI-driven product visual search and embedding comparison	Personalized recommendations using content-based filtering	Focused solely on clothing products

Despite the growing popularity of these technologies, existing platforms often lack the seamless integration of visual search and personalized recommendations. For example, while platforms like Amazon provide advanced recommendation systems, their visual search capabilities are limited in scope. Similarly, dedicated visual search

apps like Pinterest excel in image-based search but are focused less on providing actual products.

The proposed "VisuoFind" system aims to bridge these gaps by integrating AI-powered visual search and content based recommendation functionalities. The system uses advanced machine learning models for product recognition and recommendation, creating a unified platform that enhances the shopping experience. By combining the strengths of content filtering, deep learning, and computer vision, the proposed system delivers a holistic solution that addresses the limitations of existing applications. Our project does not aim to introduce entirely new features; instead, it aims to integrate all of these existing features into a single platform.

CHAPTER 3

TOOLS AND METHODOLOGY

3.1 TOOLS

The following tools were used for the development of the system:

- **Spring Boot:** A Java-based framework used to build the backend of the application, enabling rapid development of RESTful APIs and seamless integration with various services.
- **React.js:** A JavaScript library used to build the user interface of the application.
- **Tailwind CSS:** A utility-first CSS framework used to design and style the frontend of the application.
- **PostgreSQL:** A relational database system used to store and manage application data.
- **Python:** Used for implementing machine learning models and AI functionalities, including:
 - **Feature Extraction:** Implemented using ResNet-50 to extract visual features from uploaded images, and SBERT to extract semantic features from product descriptions.
 - **Similarity Search:** Implemented using FAISS for efficient product matching based on image and text.
 - **Content-Based Recommendation:**
 - **Image-Based Similarity:** Utilizes ResNet embeddings with cosine or Euclidean distance for similarity scoring.
 - **Text-Based Similarity:** Uses SBERT to perform semantic similarity matching between product descriptions and user queries.
- **Git:** A version control system used to track changes in the source code and facilitate team collaboration.
- **GitHub:** A web-based platform used to host the project repository, manage version control, support collaboration, and issue tracking.

3.2 DATABASE DETAILS

PostgreSQL has been utilized for data storage and management in the VisuoFind system. It is a powerful, open-source relational database that supports structured data, data integrity, indexing, and advanced querying capabilities. VisuoFind uses PostgreSQL to store and retrieve various types of information including user profiles, product metadata, search logs, image and text embeddings, and recommendation history. PostgreSQL integrates with the Java Spring Boot backend, providing a robust and efficient interface for managing relational data operations.

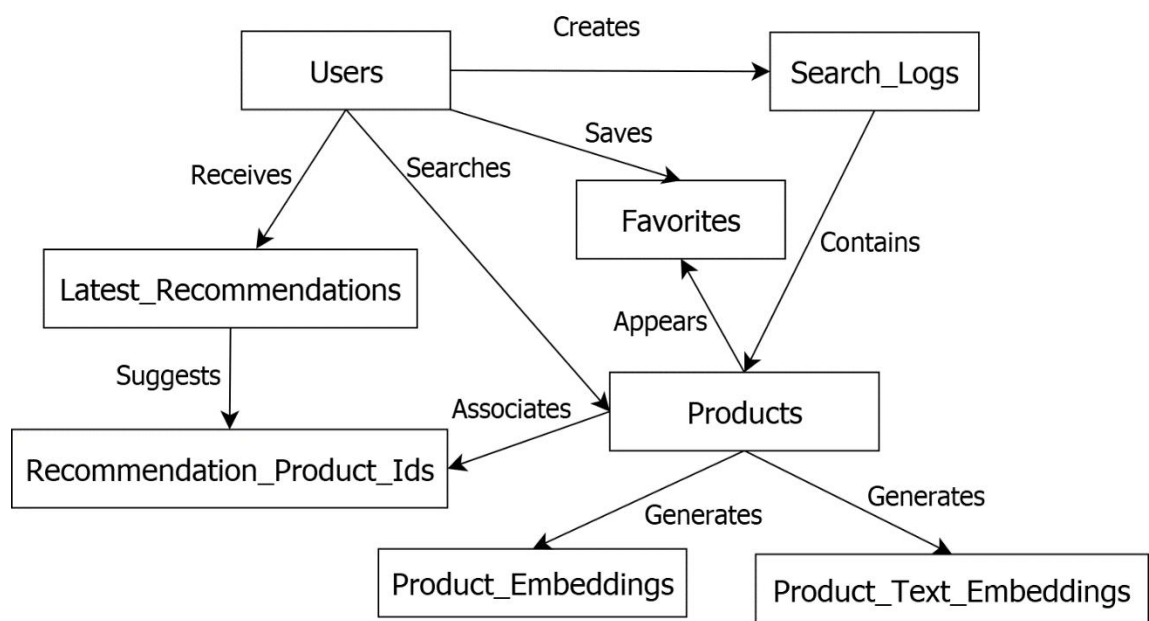


Figure 3.1: Block Diagram of Database Collections

3.2.1 DATABASE SCHEMA

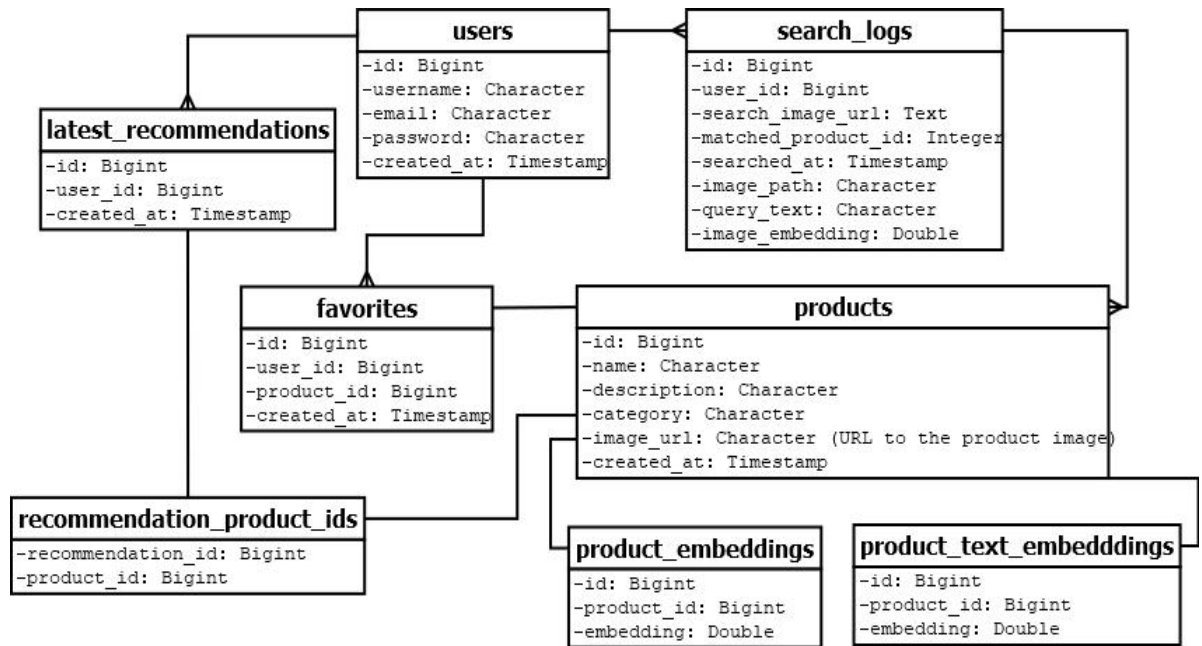


Figure 3.2: Database Schema of VisuoFind

The database schema of VisuoFind is designed to support image-based product search and content-based recommendation functionalities. It comprises several relational tables that manage user data, product metadata, embedding vectors, and user interaction logs. Below is an overview of the key tables in schema, along with their descriptions, and their primary fields:

1. Users:

Stores information about users registered on the VisuoFind platform.

- **Key Fields:**

- id (*Primary Key*): Unique identifier for each user.
- username: Username chosen by the user.
- email: User's email address.
- password: Securely hashed password.
- created_at: Timestamp indicating when the account was created.

2. Products:

Contains metadata for all products available on the platform.

- **Key Fields:**

- *id (Primary Key)*: Unique identifier for each product.
- *name*: Name of the product.
- *description*: Detailed description of the product.
- *category*: Category/type of the product (e.g., clothing, accessories).
- *image_url*: URL pointing to the product image.
- *created_at*: Timestamp when the product was added.

3. Product_Embeddings:

Stores image embeddings used for visual similarity search.

- **Key Fields:**

- *id (Primary Key)*: Unique identifier for each embedding record.
- *product_id*: Foreign key referencing the associated product.
- *embedding*: Vector representation of the product image.

4. Product_Text_Embeddings:

Stores text embeddings of product descriptions for semantic search.

- **Key Fields:**

- *id (Primary Key)*: Unique identifier for each text embedding.
- *product_id*: Foreign key referring to the associated product.
- *embedding*: Vector representation of the product text.

5. Search_Logs:

Logs user search activity, including the search queries, and matched results.

- **Key Fields:**

- *id (Primary Key)*: Unique identifier for each search log entry.
- *user_id*: Foreign key referencing the user who performed the search.
- *search_image_url*: URL of the uploaded search image (if applicable).
- *image_path*: Local or server-side path to the image.
- *query_text*: Optional text query provided by the user.
- *matched_product_id*: Foreign key referencing the product matched.
- *image_embedding*: Embedding vector used during the search.
- *searched_at*: Timestamp indicating when the search was performed.

6. Favorites:

Stores information about products favorited (saved) by users.

- **Key Fields:**

- *id (Primary Key)*: Unique identifier for each favorite record.
- *user_id*: Foreign key referencing the user who favorited the search.
- *product_id*: Foreign key referencing the favorited product.
- *created_at*: Timestamp indicating when the product was added to favorites.

7. Latest_Recommendations:

Maintains tags associated with products to improve categorization and search.

- **Key Fields:**

- *id (Primary Key)*: Unique identifier for each recommendation entry.
- *user_id*: Foreign key referencing the user receiving recommendation.
- *created_at*: Timestamp indicating when the recommendation was generated.

8. Recommendation_Product_Ids:

Stores the mapping between recommendations and the products included in them.

- **Key Fields:**

- *recommendation_id*: Foreign key referencing the recommendation.
- *product_id*: Foreign key referencing the recommended product.

3.3 METHODOLOGY

In modern computer vision and machine learning, reverse image search is commonly referred to as Content-Based Image Retrieval (CBIR). Unlike traditional search engines that rely on textual metadata, CBIR systems retrieve images based on their visual characteristics, such as color, shape, texture, and spatial patterns. This allows for much more intuitive and flexible image searches without need for manual tagging or labeling.

For VisuoFind, the objective is to develop an AI-powered visual search and recommendation system that combines deep learning models (Convolutional Neural Networks like ResNet-50), K-Nearest Neighbors (KNN) and FAISS for efficient vector indexing, and text similarity techniques (SBERT) for semantic understanding and similarity in textual descriptions.

1. Feature Extraction using CNN Architecture:

The system begins with a Convolutional Neural Network (CNN) that converts the input image into a feature vector, a high-dimensional numerical representation of its visual properties. CNNs are well-suited for this task due to their layered architecture, which captures various abstraction levels:

- **Low-level features:** Colors, textures, basic shapes.
- **Mid-level features:** Object parts, spatial arrangements.
- **High-level features:** Semantic concepts such as product category or usage patterns.

VisuoFind employs ResNet-50, a 50-layer deep CNN pre-trained on the ImageNet dataset. Each image is resized to $224 \times 224 \times 3$, reshaped to a batch format of (1, 224, 224, 3), and then passed through the model to generate a 1000-dimensional embedding. These embeddings are flattened and normalized to standardize scale, improving comparison accuracy.

The CNN's translation invariance, enabled by convolutional filters and pooling layers, ensures consistent detection of key features regardless of object position, orientation, or lighting conditions.

2. Pooling and Dimensionality Reduction:

Within the CNN architecture, Max Pooling layers follow convolutional layers to reduce the spatial dimensions of the feature maps. This reduces computational load and improves generalization. Benefits include:

- Faster inference by reducing the size of feature maps
- Focused attention on the most relevant visual areas
- Resilience to image noise, background clutter, and minor distortions

Pooling enables the system to concentrate on key visual patterns such as edges, textures, and symmetry which are particularly critical in recognizing products.

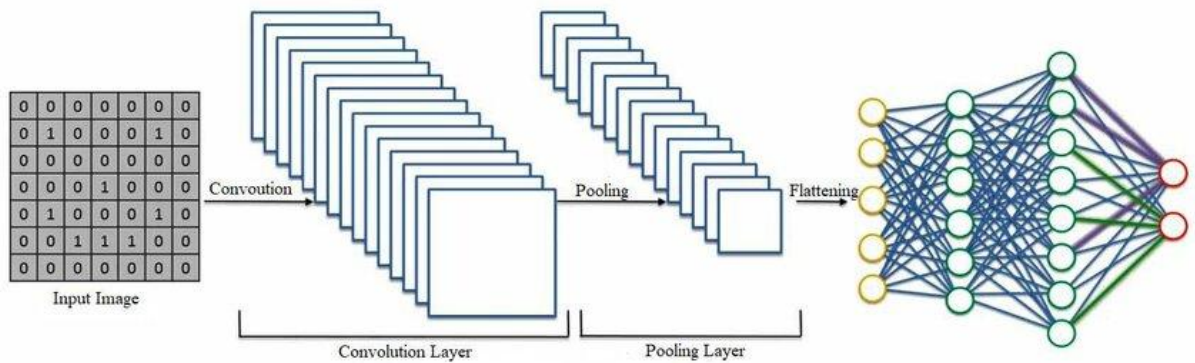


Figure 3.3: Core Operation of a CNN Network

3. Feature Indexing and Similarity Search with KNN and FAISS:

After generating feature vectors for all product images, the system uses FAISS to index them. FAISS is a library optimized for fast similarity search in high-dimensional vector spaces.

When a user uploads an image:

- The image is passed through ResNet-50 to generate its embedding.
- FAISS performs K-Nearest Neighbor (KNN) search in the embedding space.
- Euclidean distance is used as similarity metric to find the most visually similar products.

This approach ensures that product matches are based on deep visual features, not superficial pixel-level comparisons.

4. Visual Search Workflow:

The complete visual search pipeline in VisuoFind consists of the following steps:

Step 1: User Uploads an Image:

The user uploads an image via the web interface. This image can be a photo or a saved product image.

Step 2: Image Preprocessing:

Once uploaded, the image undergoes preprocessing to ensure compatibility:

- Loaded using the Pillow (PIL) library.
- Resized to a standard input shape (224×224×3) that the ResNet-50 model expects.
- Converted into a batch format: a 4D tensor with shape (1, 224, 224, 3) for inference with Keras.

Step 3: Feature Extraction Using CNN:

The preprocessed image is passed through the ResNet-50 model to generate a 1000-dimensional feature embedding. This embedding captures the visual characteristics of the image, such as color gradients, patterns, edges, and abstract shapes. The resulting vector is then flattened and normalized, to ensure all vectors are on a common scale, which is crucial for accurate comparison.

Step 4: Image Vector Comparison with KNN + FAISS:

The normalized embedding is compared against the indexed product vectors using KNN search via FAISS. It identifies the closest matches based on Euclidean distance as the metric.

Step 5: Return and Display Results:

The system returns the top visually similar products based on KNN results. These images represent products from the database that share key visual features with the uploaded image. This similarity is robust to common visual variations like:

- Different backgrounds
- Slight rotations or translations
- Changes in scale or lighting

Finally, the selected images are displayed on the frontend to the user. The user sees a side-by-side comparison between their uploaded image and the recommended products.

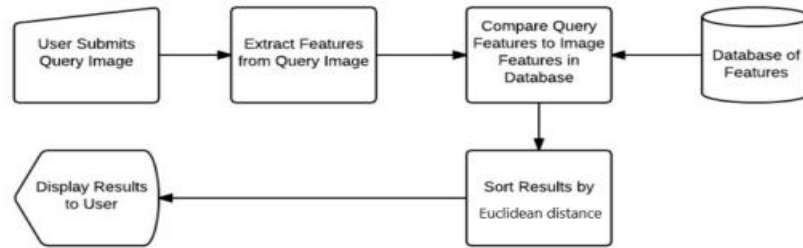


Figure 3.4: Block Diagram of Visual Search Workflow

5. Text-Based Similarity for Hybrid Searches:

In addition to visual similarity, VisuoFind incorporates text-based similarity to provide hybrid results that are both visually and contextually relevant. Each product in the database includes a title and description which are processed using SBERT (Sentence-BERT) to generate semantic embeddings to capture the meaning behind the texts. During hybrid recommendation:

- The user’s query is compared against other product descriptions.
- Cosine similarity is computed between SBERT-generated text embeddings.
- Scores from both visual similarity and textual similarity are combined to combined to rank results.

This hybrid approach enhances search quality, especially when visual features are minimal or ambiguous (e.g., a plain black shirt) or searches are for abstract categories (e.g., “casual evening wear”).

6. Model Export and Reusability:

After feature extraction and indexing:

- All 1000-D feature vectors are serialized and saved using Pickle module.
- These saved embeddings allow instant retrieval during user searches without having to reprocess the dataset each time.
- This design significantly improves response time and allows the system to scale efficiently.

7. Building the Web Application (Frontend + Backend):

- **Frontend:** Built using React.js, the UI enables users to upload images, enter queries, and view matched products in real time.
- **Backend:** Developed using Spring Boot and Python (FastAPI) handles image processing, feature extraction, vector indexing, and similarity search.

The VisuoFind web application integrates deep learning, KNN-based similarity search, and hybrid recommendation techniques to streamline product discovery and deliver a personalized shopping experience. By learning from user interactions and search behavior, the system adapts to individual preferences, offering accurate, visually relevant suggestions that enhance overall experience in e-commerce platforms.

3.4 APPROACH USED

For the development of VisuoFind, we adopted a hybrid approach that combines aspects of both the waterfall and iterative-incremental models. Initially, we followed a structured process typical of the waterfall model, starting with requirement gathering, followed by design, development, and testing. (Lutkevich, 2006).

However, given the experimental nature of integrating AI models and refining recommendation logic, we introduced iterative cycles into our workflow. This allowed us to continuously test and improve core features like visual search and content-based recommendations, adapt to changing requirements, and incorporate new techniques as the project evolved.

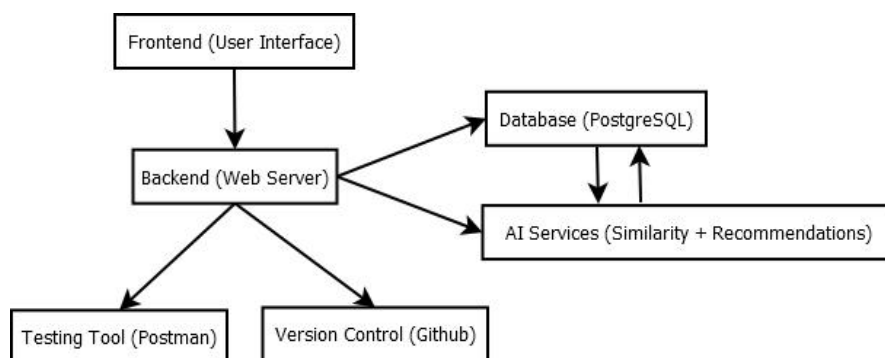


Figure 3.5 : Block Diagram of System Architecture

The block diagram is designed to provide a high-level overview of the architecture, focusing on the interaction between the key components of the VisuoFind system.

Frontend (React.js + CSS):

- User-friendly interface for uploading images and viewing visual matches.
- Displays personalized recommendations based on user interactions.
- Includes authentication, browsing history tracking, and responsive UI.

Backend (Java Spring Boot):

- Manages API routing, authentication, session management, and request delegation.
- Acts as a bridge between the frontend, AI services, and database.

AI Services (Python-based Microservices):

- **Image Feature Extraction:** Uses ResNet-50 CNN to generate feature vectors from uploaded images.
- **Visual Search (FAISS):** Performs high-speed similarity search on image embeddings using FAISS.
- **Text Similarity (SBERT):** Extracts semantic embeddings from product descriptions and user queries.
- **Recommendation Engine:** Combines visual and text-based similarities for personalized recommendations.
- **KNN Classifier:** Adds an efficient classification layer to group products based on embedding space clusters.

Database (PostgreSQL):

- Stores user profiles, product metadata, embedding vectors, and search log.

Testing Tool:

- Postman used to test RESTful APIs for ensuring seamless integration and communication across services.

Version Control:

- GitHub for code versioning, collaborative development, and modular project structure.

3.5 DESIGN

3.5.1 ENTITY RELATIONSHIP DIAGRAM

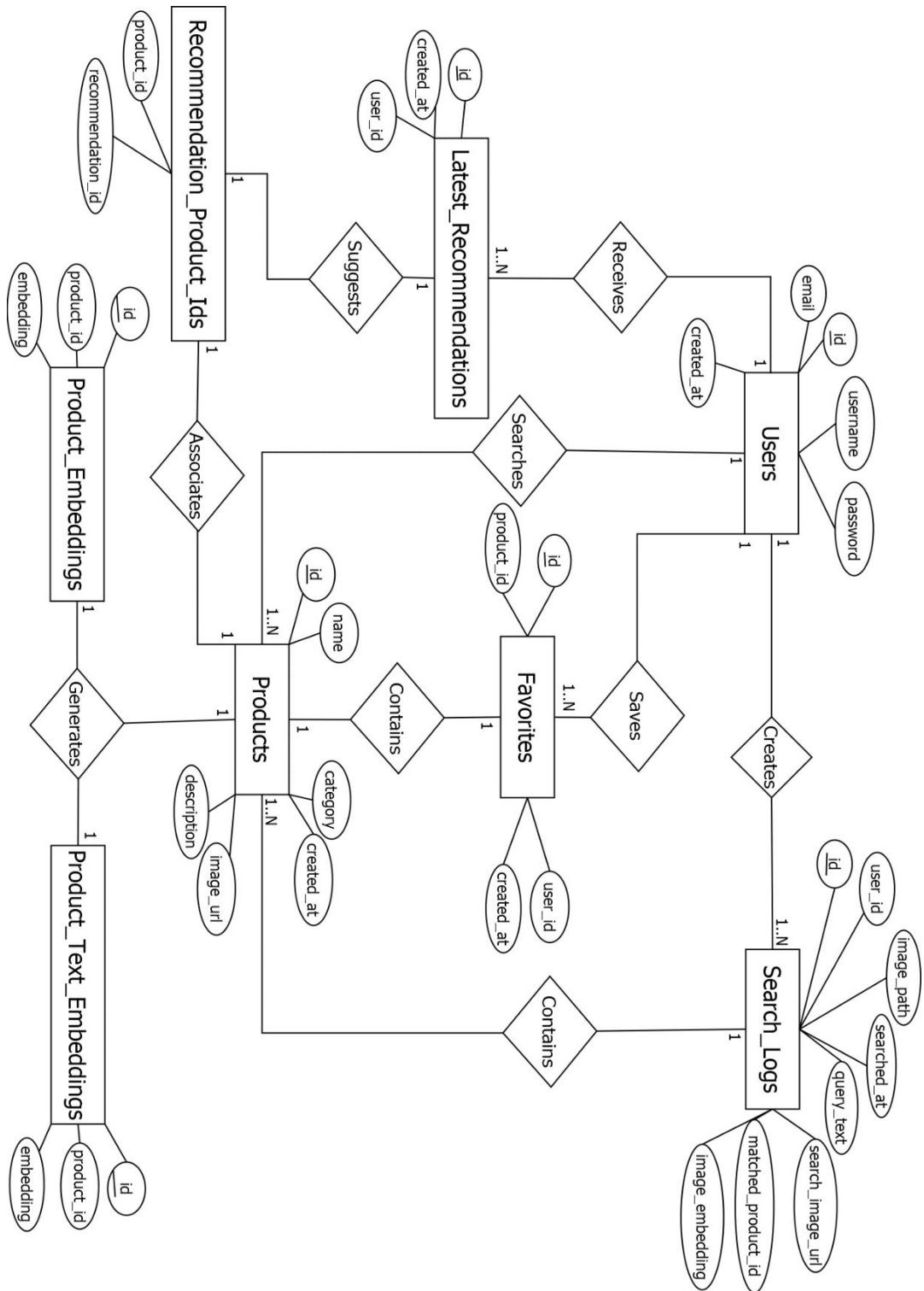


Figure 3.6: Entity Relationship Diagram of VisuoFind

3.5.2 USE CASE DIAGRAM

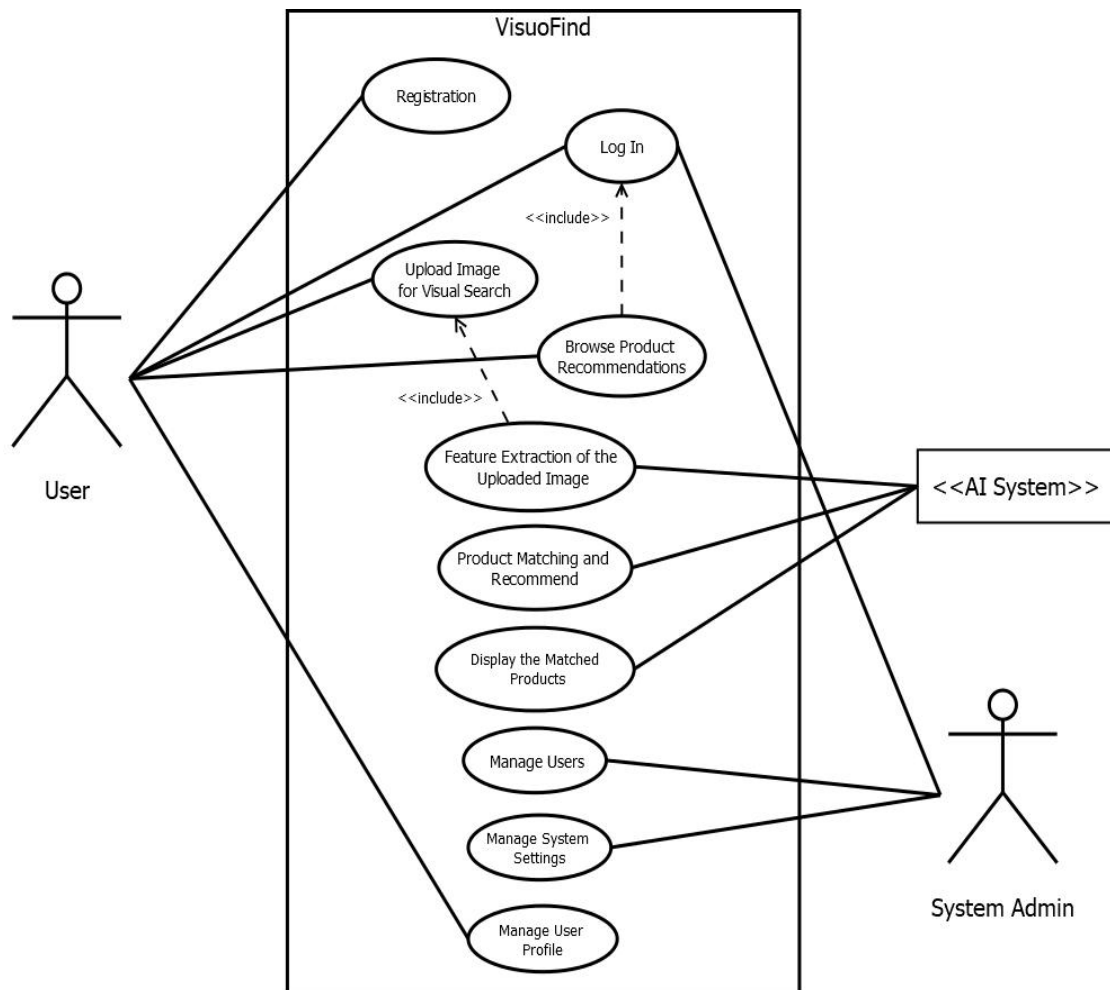


Figure 3.7 : Use Case Diagram for VisuoFind

This use case diagram illustrates the primary actions users can perform in VisuoFind, such as upload image for visual search, feature extraction of uploaded image, display recommendations.

Use Case UC 1: Upload Image for Visual Search

Primary Actor: User

Stakeholders and Interests:

- User: Wants to find a similar product by uploading an image of an item they are interested in.
- System: Ensures the image is processed correctly for visual search and feature extraction.

Pre-Conditions:

- User must be logged in to the system.
- User has an image file of the product they want to search for.

Post-Conditions:

- The image is uploaded and cropped to focus on the product itself.
- The system prepares the image for feature extraction and product matching.

Main Success Scenario:

- The user navigates to the visual search section of the website.
- The system displays an option to upload an image.
- The user selects an image file from their device.
- The system displays a preview of the uploaded image.
- The user clicks the “Upload” button to submit the image.
- The system crops the image to focus on the product itself.
- The image is sent to the system for feature extraction and processing.

Alternative Scenario:

- If the uploaded image is in an unsupported format, the system prompts the user to upload a different file format (JPEG, PNG).
- If the user cancels the upload process, the image is discarded, and no further action occurs.

Use Case UC 2: Feature Extraction of Uploaded Image

Primary Actor: System

Stakeholders and Interests:

- User: Expects the system to accurately identify features of the uploaded image to make effective product recommendations.
- System: Needs to process the image and extract visual features to generate relevant product recommendations.

Pre-Condition:

- The image has been uploaded and cropped successfully.
- The system has access to pre-trained Convolutional Neural Network (CNN) models (such as ResNet-50) for feature extraction.

Post-Condition:

- The system extracts features (such as shape, color, and style) from the image.
- The extracted features are stored and ready for product matching and recommendation.

Main Success Scenario:

1. After the user uploads and crops the image, the system sends the image to the AI model for processing.
2. The AI model applies a pre-trained Convolutional Neural Network (CNN) specifically ResNet-50, to the image.
3. The CNN analyzes the image for key visual features such as color, and shape.
4. The extracted features are stored in the system's feature database.
5. The system prepares the image data for the product matching phase.

Alternative Scenario:

- If the feature extraction model fails or encounters errors, the system retries the extraction process.
- If an error persists, the system prompts the user to upload a new image.

Use Case UC 3: Display for Recommendations

Primary Actor: User

Stakeholders and Interests:

- User: Wants to receive personalized product recommendations based on visual search and past browsing behavior.
- System: Delivers relevant and personalized recommendations by analyzing user history.

Pre-Condition:

- The image has been processed, and features have been extracted.
- The user is logged in and has a browsing history that the system can analyze. The system has access to user data, including past searches, clicks, and purchases.

Post-Condition:

- Recommendations are displayed, combining visual search results with personalized suggestions from the user's browsing history.

Main Success Scenario:

- After image feature extraction, the system uses K-Nearest Neighbors (KNN) for visual search results.
- The system analyzes the user's browsing history (e.g., viewed products, favorited products).
- Content Filtering is applied to create personalized recommendations based on the user's past interactions, preferences, and browsing behavior.
- Recommended products are displayed, mixing visually similar items with those aligned to user preferences.

Alternative Scenario:

- If no browsing history is found (new user), generic recommendations based on product features and image similarity are shown.
- If browsing behavior is diverse or not sufficient, the system combines multiple recommendation types (popular items, related products).

3.5.3 SYSTEM SEQUENCE DIAGRAM

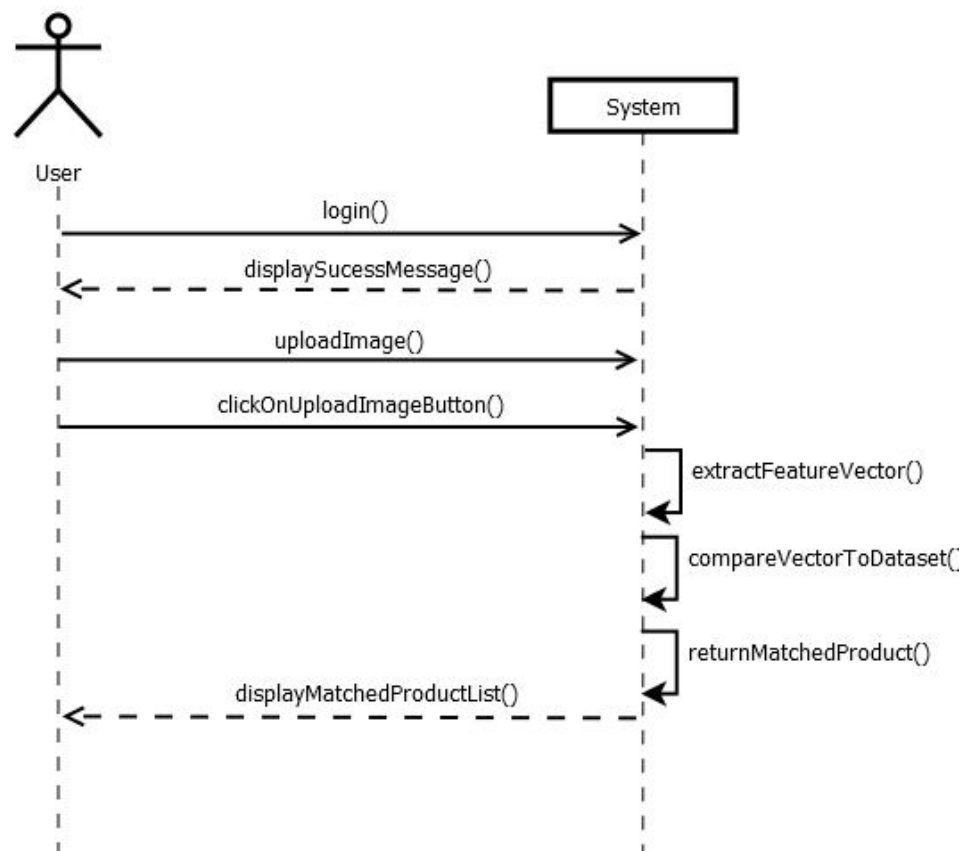


Figure 3.8: System Sequence Diagram for Upload Image for Visual Search

This diagram shows the sequence of events that occur when user uploads an image for visual search operation.

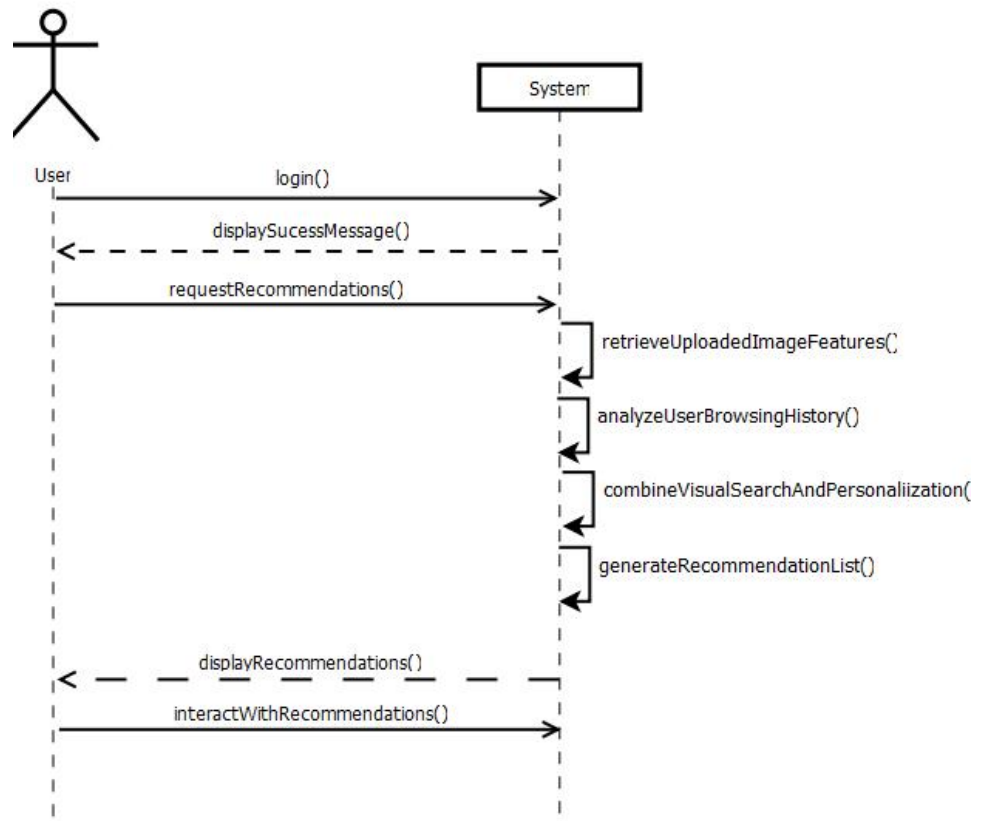


Figure 3.9: System Sequence Diagram for Display Recommendation for User

This diagram illustrates the sequence for the system to generate the recommendation list for the user.

3.5.4 DESIGN CLASS DIAGRAM

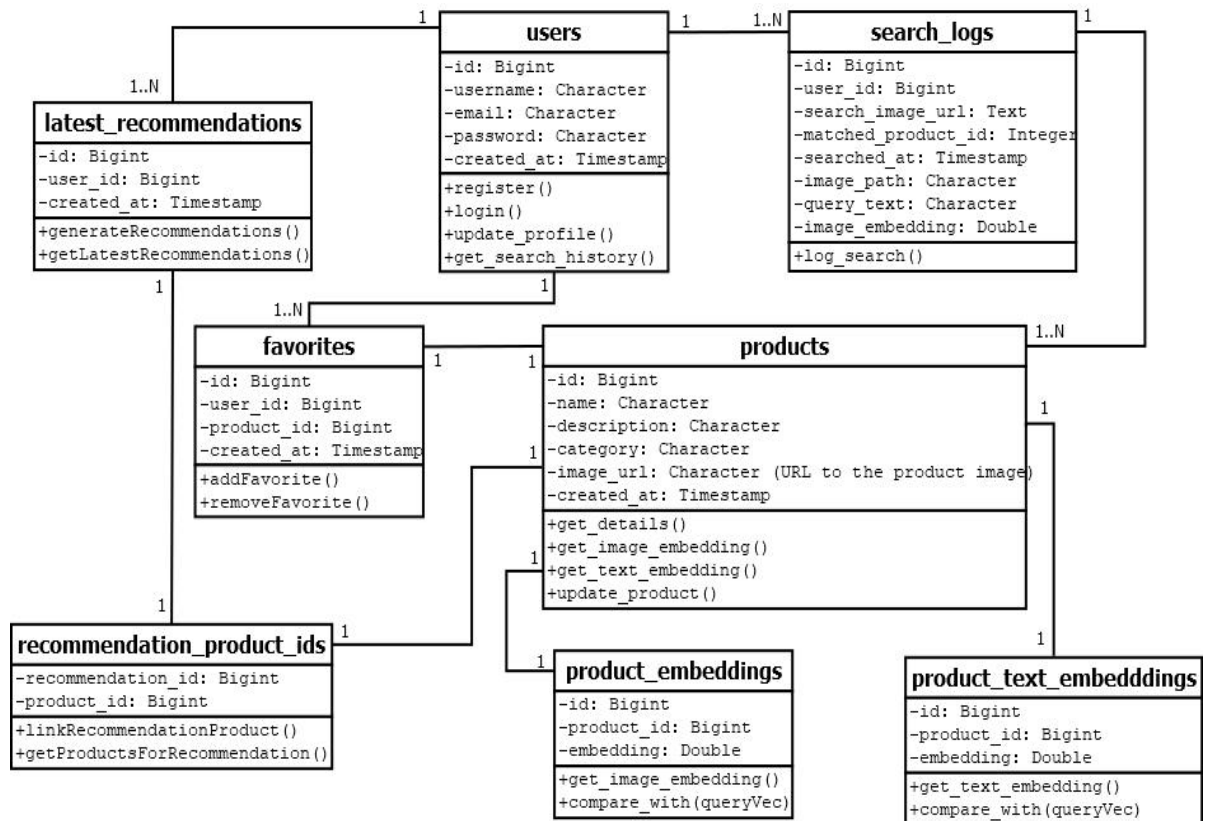


Figure 3.10: Design Class Diagram of VisuoFind

The design class diagram provides an overview of the key classes within the VisuoFind system, including products, users, orders, cart, etc. It also shows their attributes and relationships with each other.

3.5.5 INTERACTION DIAGRAM

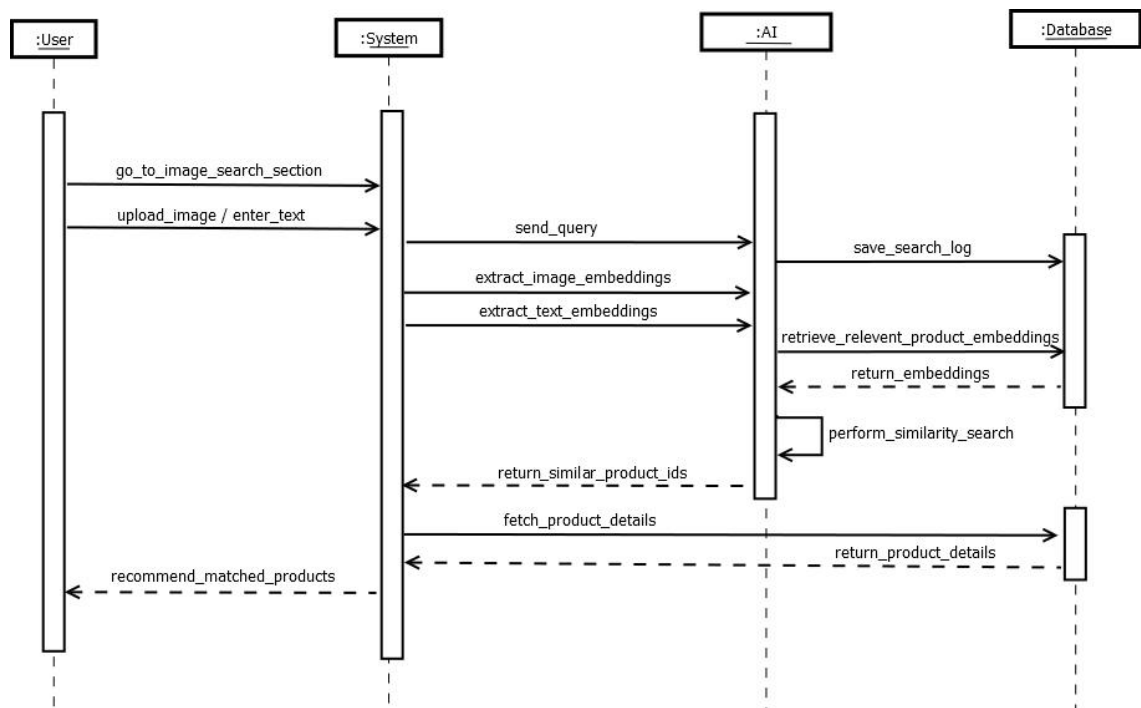


Figure 3.11: Interaction Diagram for Visual Search of VisuoFind

This interaction diagram illustrates how the system processes an image or text search, showing the flow of operation between the user, system, AI, and database.

CHAPTER 4

TEST CASES

4.1 SOFTWARE TESTING

Software testing is a process to evaluate the functionality of a software application with the intent to verify whether the developed software meets the specified requirements. It also identifies defects to ensure the product is reliable, accurate, and provides a quality user experience.

4.2 TEST OBJECTIVES

The main objectives of testing VisuoFind were to:

- Verify that the visual search system accurately finds and displays similar products based on uploaded query images.
- Ensure the application functions smoothly and reliably, handling user inputs and displaying results without errors or delays.

4.3 TEST CASE REPORT

4.3.1 DATASET AND SETUP

The dataset contains approximately 1000 fashion product images, with 900 images used for training embedding models and 100 images reserved for testing. Images were resized uniformly to 224×224 pixels. For focused embedding extraction, images were cropped around the main product using bounding box techniques to ensure embeddings from ResNet50 represent the product accurately.

The dataset covers multiple fashion product categories such as shoes, topwear, hoodies, watches, bags, and more. Each category was tested with 10 different query images, mostly external and unseen during training, to simulate realistic user search scenarios.

4.3.2 TEST APPROACHES AND RESULTS

After preprocessing, multiple approaches were implemented and evaluated to determine the effectiveness of visual and text-based retrieval. The results and analysis of each approach are detailed below.

Approach 1: Visual Search using ResNet50 + FAISS:

This approach uses ResNet50 (pretrained on ImageNet) for generating deep image embeddings. Each image is passed through ResNet50 (excluding the top layer) with Global Average Pooling, producing a 1000-dimensional vector.

- **Embedding Generation:** ResNet50
- **Indexing:** FAISS
- **Search Process:** User-uploaded image is embedded and compared to indexed embeddings to return top-N similar products.

Results:

- **Top-5 Retrieval Accuracy:** Up to 85% in controlled evaluations
- **Average Search Time:** <300 ms observed
- **Image Embedding Dimensions:** 1000

Insights:

- Performed exceptionally well for visually distinct products such as colorful clothing, or unique patterns.
- Works best when user queries are based on clear visual cues (e.g., uploading a photo).
- Struggled with semantic similarity; items that look different but serve the same purpose were ranked lower.

Approach 2: Text Search using SBERT:

This module supports text-based product retrieval using SBERT (Sentence-BERT) embeddings. Each product description is embedded into a dense vector space, and cosine similarity is used to compare user queries with these embeddings.

- **Text Embedding:** SBERT
- **Similarity Metric:** Cosine similarity
- **Search Method:** Top-N product descriptions ranked by closeness to query vector.

Results:

- **Top-5 Retrieval Accuracy:** 80%
- **Latency:** ~250 ms/query
- **Text Embedding Dimension:** 384

Insights:

- SBERT handled natural language queries effectively, capturing both syntactic and semantic meaning.
- Returned highly relevant results even when user input was vague or contained synonyms (e.g., "things to check time").
- Although not used as a standalone feature, SBERT embeddings significantly enhanced the hybrid search, especially when visual cues were ambiguous or insufficient.

Approach 3: Hybrid Recommendation (Visual + Text):

This approach combines visual embeddings (from ResNet50) and textual embeddings (from SBERT) to improve the accuracy and relevance of product retrieval. Each product has both image and text embeddings. A weighted similarity score is computed by combining the cosine similarity from both modalities.

Hybrid Similarity Score = $(0.6 \times \text{Visual Similarity}) + (0.4 \times \text{Text Similarity})$

- **Visual Embedding:** ResNet50 (1000-d)
- **Text Embedding:** SBERT (384-d)
- **Combined Search:** Cosine similarity scores are weighted and summed to rank top-N results.

Results:

- **Precision@5:** Up to 91% in controlled evaluations
- **Average Response Time:** ~300-400 ms observed during testing

Insights:

- Outperformed both visual-only and text-only approaches in terms of retrieval precision.
- Especially effective when both the image and query description provided complementary context, improving disambiguation (e.g., distinguishing “formal white shirt” from “casual white shirt”).
- Helped bridge the gap when either the image or the description was unclear or partial, enhancing robustness of search.

Approach 4: Recommendation Engine (Content-Based Filtering):

This approach delivers personalized product recommendations using Content-Based Filtering (CBF). It leverages the user's past interaction history (e.g., previously viewed or favorited products) and compares them with the rest of the catalog using visual (ResNet50) and textual (SBERT) embeddings.

- **User Profile Vector:** Mean pooled embedding of user's interacted products (image + text)
- **Recommendation Process:** Nearest neighbors are retrieved using cosine similarity via FAISS
- **Components Used:** ResNet50 + SBERT (same hybrid embedding strategy as in search)

Results:

- **Top-3 Suggestion Accuracy:** 82%
- **Cold Start Strategy:** Default fallback to top-ranked hybrid clusters
- **Latency:** ~350 ms on average

Insights:

- Provided highly relevant recommendations when users had at least 5 prior interactions, with quality improving as interaction depth increased.
- Users with visually and semantically diverse browsing history received more diverse recommendations.
- Cold-start users still received meaningful results via popular cluster similarity, maintaining system robustness.

4.3.3 SAMPLE SEARCH EXAMPLE (SHOES CATEGORY):

To simulate a typical user query, a shoe image was used as input. The system extracted image embeddings and performed retrieval using Euclidean distance. The top-5 returned results included visually similar shoes with matching color, shape, and style.

Incorporating SBERT text embeddings refined rankings further, especially when product descriptions were leveraged. This hybrid-search effectively handled cases with ambiguous visual cues.

4.3.4 RETRIEVAL ACCURACY TABLES:

Each category was evaluated with 10 distinct query images, and retrieval accuracy was calculated as the percentage of relevant matches within the top-5 results.

Table 4.1: Table of Results Using Euclidean Distance for Image Embedding

Category	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6	Sample 7	Sample 8	Sample 9	Sample 10	Average
Shoes	1	0.9	0.9	0.8	0.9	0.9	1	0.8	0.8	0.8	0.88
Tops	0.9	0.9	0.8	0.9	1	0.9	0.9	0.9	0.8	0.9	0.9
Bags	0.8	0.7	0.8	0.7	0.8	0.8	0.7	0.8	0.7	0.7	0.75
Watches	1	0.9	1	0.9	1	1	1	1	0.9	1	0.97
Jeans	0.7	0.6	0.7	0.7	0.7	0.6	0.7	0.7	0.6	0.7	0.67
Dresses	0.8	0.9	0.9	0.8	0.8	0.9	0.8	0.8	0.9	0.8	0.84
Shirts	0.9	0.8	0.9	0.9	1	0.8	0.9	1	0.9	0.9	0.91
Sandals	0.7	0.6	0.6	0.7	0.7	0.6	0.7	0.6	0.7	0.6	0.65
Jackets	0.8	0.9	0.8	0.9	0.8	0.8	0.9	0.8	0.8	0.9	0.85
T-shirts	1	0.9	1	0.9	0.9	1	0.9	0.9	1	0.9	0.94

Table 4.2: Table of Results Using Cosine Similarity for Text Embedding

Category	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6	Sample 7	Sample 8	Sample 9	Sample 10	Average
Shoes	0.8	0.7	0.7	0.7	0.8	0.7	0.8	0.7	0.7	0.7	0.73
Tops	0.9	0.8	0.8	0.9	0.9	0.9	0.8	0.8	0.9	0.9	0.87
Bags	0.6	0.7	0.7	0.7	0.7	0.6	0.7	0.6	0.7	0.7	0.67
Watches	0.9	0.8	0.9	0.8	0.9	0.9	0.9	0.9	0.8	0.9	0.86
Jeans	0.7	0.6	0.6	0.6	0.7	0.6	0.6	0.6	0.6	0.6	0.62
Dresses	0.8	0.9	0.9	0.8	0.8	0.9	0.8	0.8	0.9	0.8	0.84
Shirts	0.8	0.8	0.8	0.9	0.9	0.8	0.8	0.9	0.8	0.9	0.85
Sandals	0.6	0.5	0.5	0.6	0.6	0.5	0.6	0.5	0.5	0.6	0.55
Jackets	0.8	0.7	0.7	0.8	0.8	0.7	0.7	0.7	0.7	0.8	0.74
T-shirts	0.9	0.9	0.9	0.9	0.8	0.9	0.9	0.8	0.9	0.9	0.88

Table 4.3: Table of Results for Hybrid Embedding

Category	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6	Sample 7	Sample 8	Sample 9	Sample 10	Average
Shoes	0.9	0.8	0.9	0.9	0.9	0.8	0.9	0.8	0.9	0.9	0.87
Tops	0.95	0.9	0.9	0.95	0.95	0.9	0.9	0.9	0.95	0.9	0.92
Bags	0.75	0.75	0.75	0.8	0.8	0.75	0.75	0.8	0.8	0.8	0.77
Watches	0.95	0.9	1	0.9	1	0.95	1	0.95	0.9	1	0.96
Jeans	0.7	0.7	0.7	0.7	0.75	0.7	0.75	0.7	0.75	0.7	0.72
Dresses	0.85	0.9	0.9	0.85	0.9	0.85	0.9	0.85	0.9	0.9	0.88
Shirts	0.9	0.9	0.9	0.9	1	0.9	0.9	0.95	0.95	0.95	0.94
Sandals	0.65	0.6	0.65	0.6	0.65	0.6	0.6	0.65	0.65	0.6	0.63
Jackets	0.85	0.85	0.85	0.85	0.85	0.9	0.85	0.85	0.85	0.9	0.86
T-shirts	0.95	0.95	1	0.95	0.95	1	0.95	1	0.95	1	0.97

4.4 TEST RESULTS

Table 4.4: Test Cases

SN	TEST CASES	EXPECTED	OBSERVED	RESULT
1	Upload Query Image	The user should be able to upload an image to be used a search query.	The system successfully accepts the image upload and generates an embedding vector.	Ok
2	Image Similarity Search	Given a query image, the system should return visually similar products/images based on embeddings.	The system returns relevant similar images ranked by similarity, matching expected results.	Ok
3	Display Search results	The search results should display product images with relevant metadata clearly and promptly.	Search results are displayed with correct images and associated information without delay.	Ok
4	Add New Products	The user should be able to add new products with their embeddings being extracted automatically.	Products are added to database with both text and image embeddings.	Ok
5	Search Bar Functionality	The user should be able to search for products using keywords in addition to image search.	Keyword search returns relevant results accurately alongside image-based search.	Ok

4.5 TEST RESULTS ILLUSTRATIONS

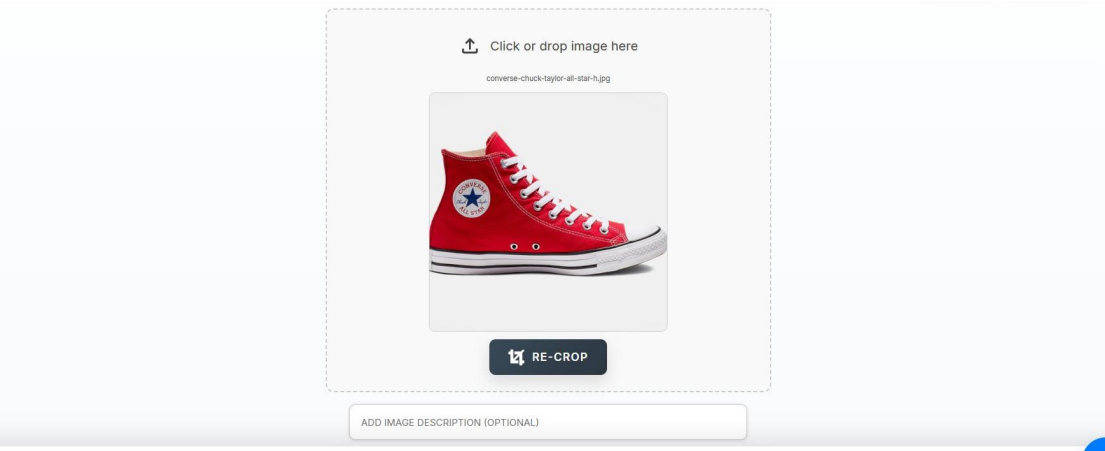


Figure 4.5: Image Upload for Visual Search

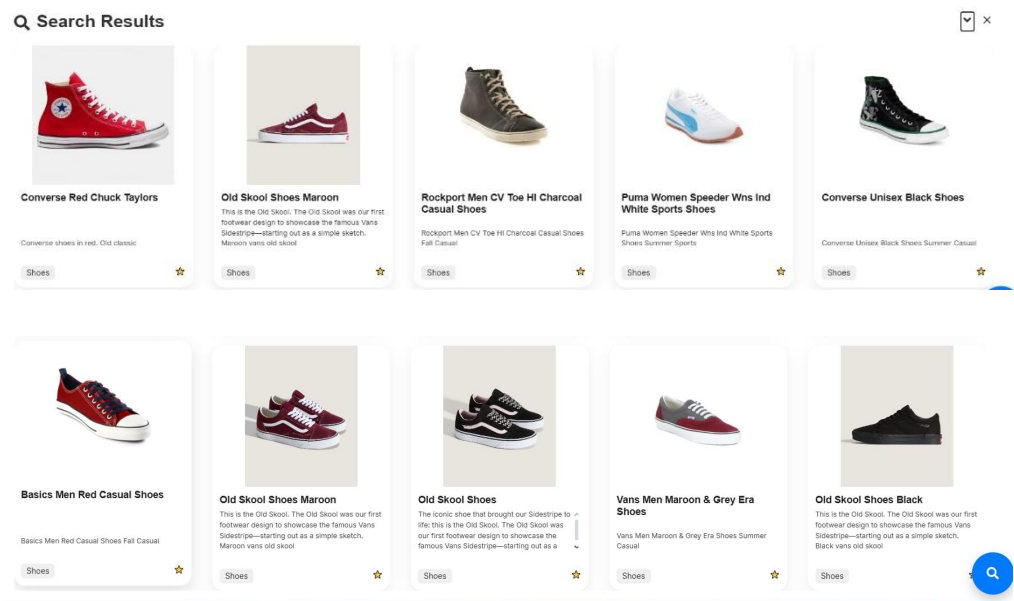


Figure 4.6: Result of Visual Search

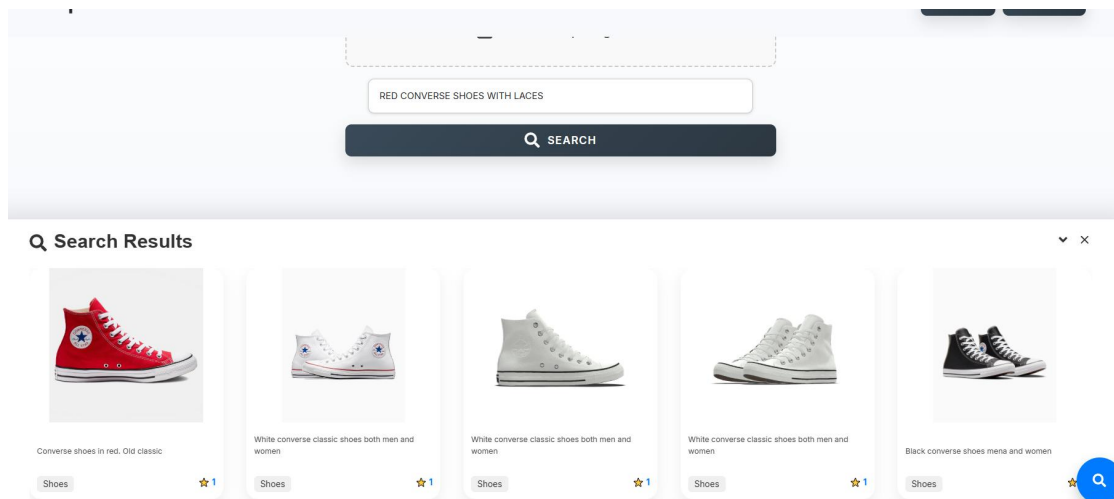


Figure 4.7: Result of Text Only Search

CHAPTER 5

CONCLUSION AND RECOMMENDATION

5.1 RECOMMENDATION

To ensure the long-term growth, effectiveness, and user satisfaction of the VisuoFind platform, the following strategic initiatives are recommended for future development cycles. These focus on advancing system intelligence, improving scalability, and enriching the user experience in product discovery and recommendation.

5.1.1 ENHANCEMENT OF VISUAL SEARCH ENGINE

The core strength of VisuoFind lies in its ability to process and understand product images. To further refine this capability, it is recommended to integrate advanced deep learning models (e.g., fine-tuned ResNet or EfficientNet) for improved feature extraction. Additionally, incorporating multimodal search that combines both image and text queries will provide users with a richer and more intuitive search experience. This enhancement would make product discovery faster, more accurate, and more engaging.

5.1.2 TRANSITION TO HYBRID RECOMMENDATION SYSTEM

Currently, VisuoFind focuses primarily on content-based filtering. For superior personalization, adopting a hybrid approach that combines content-based methods with semantic text understanding (via BERT or TF-IDF) will significantly improve recommendations. This will allow the system to capture both visual similarity and contextual meaning, ensuring that users receive more relevant and tailored product suggestions.

5.1.3 ENRICHMENT OF USER EXPERIENCE

Improving the user interface and interaction design is crucial for adoption. Recommendations include adding features such as instant visual feedback when users upload images, interactive filtering options, and personalized dashboards to track

search history. Enhancing UX will make the platform more engaging, intuitive, and accessible to users of varying technical expertise.

5.1.3 EXPANSION OF PRODUCT DATASETS

The accuracy and variety of recommendations directly depend on the breadth of available data. Expanding VisuoFind's dataset to include diverse product categories, attributes, and styles will improve the system's capability to recommend niche and trending items. Incorporating metadata such as price, popularity, and brand information will further enrich the recommendation process.

5.1.4 MOBILE-FIRST STRATEGY AND CROSS-PLATFORM SUPPORT

Since users often browse products on mobile devices, adopting a mobile-first approach is essential. Ensuring that the web application is fully responsive is the immediate step, followed by developing native mobile apps for iOS and Android. Mobile applications will provide users with additional features such as camera-based instant search, push notifications for recommendations, and offline support for cached results.

5.2 CONCLUSION

The VisuoFind system demonstrates the feasibility and effectiveness of using computer vision and content-based filtering for product search and recommendations. By leveraging ResNet for feature extraction and FAISS for similarity search, the system provides fast, accurate, and relevant results based on visual inputs. The integration of text-based similarity using TF-IDF further enhances personalization, making the platform adaptable to user preferences.

Overall, VisuoFind proves that combining image recognition with intelligent recommendation techniques creates a powerful and user-friendly solution for product discovery. With future enhancements in scalability, dataset expansion, and hybrid recommendation modeling, VisuoFind holds strong potential to become a leading platform for personalized and visually driven product search.

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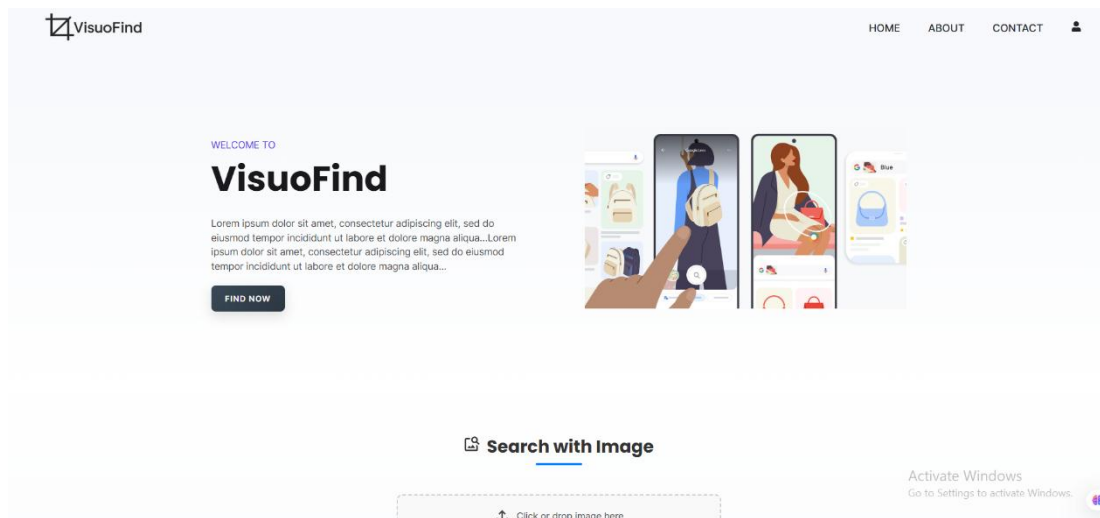
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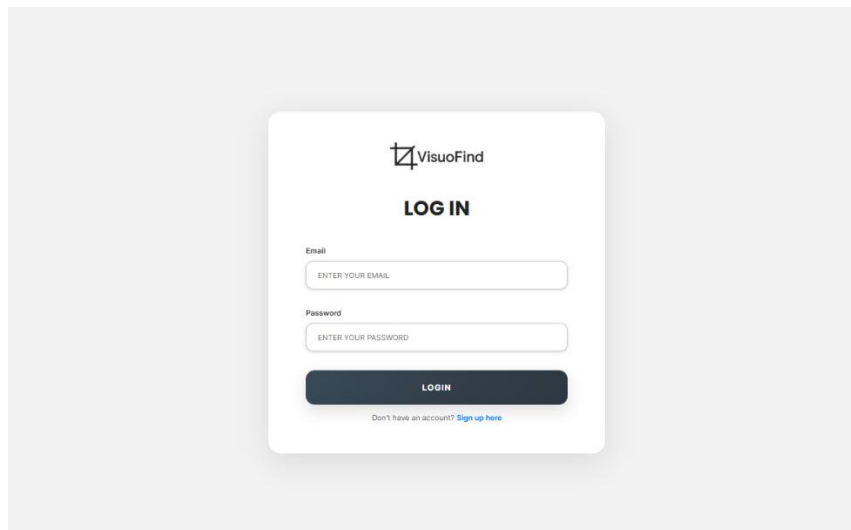
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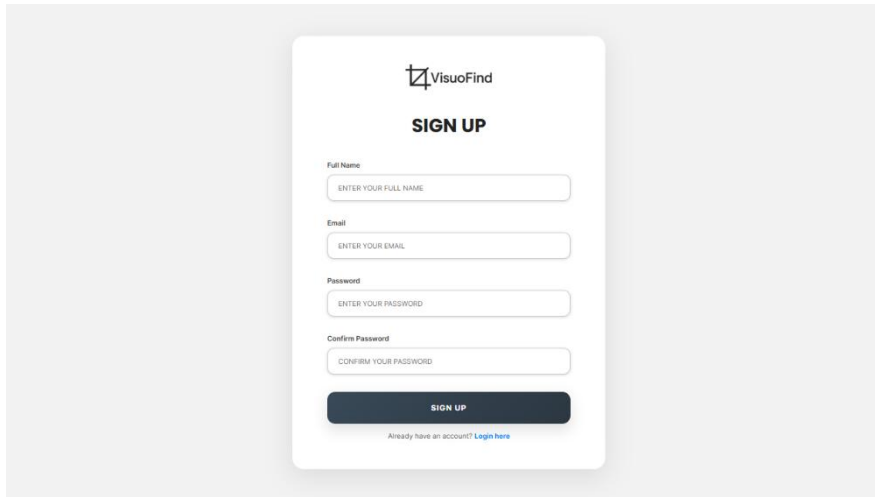
APPENDICES



Appendix 1: Home Page

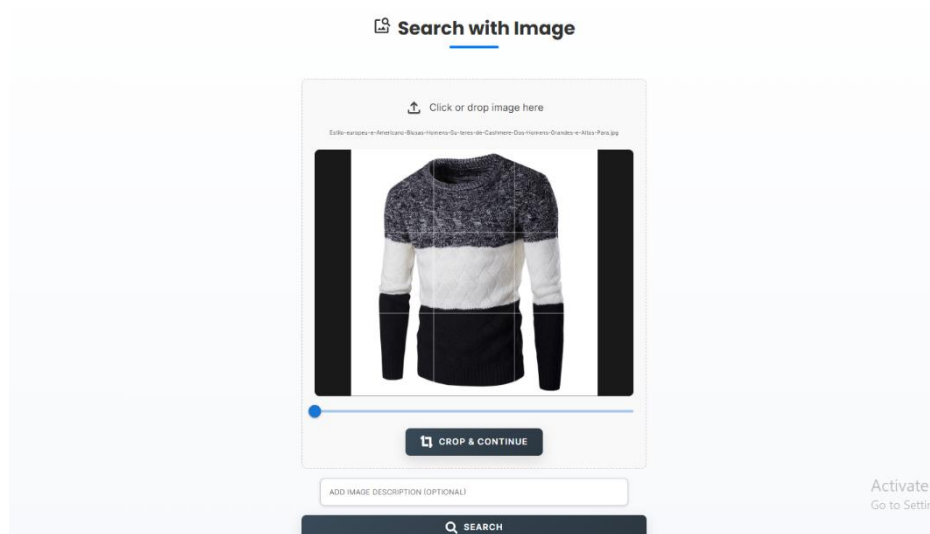


Appendix 2: Login page



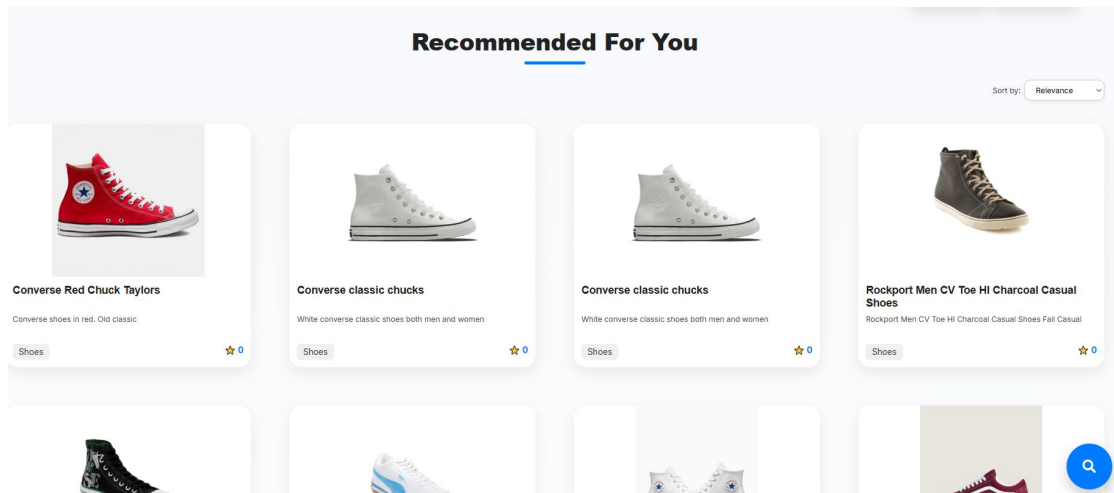
The image shows a 'SIGN UP' form for VisuoFind. At the top is the VisuoFind logo. Below it is the title 'SIGN UP'. The form contains four input fields: 'Full Name' with placeholder text 'ENTER YOUR FULL NAME', 'Email' with placeholder text 'ENTER YOUR EMAIL', 'Password' with placeholder text 'ENTER YOUR PASSWORD', and 'Confirm Password' with placeholder text 'CONFIRM YOUR PASSWORD'. Below these fields is a dark blue 'SIGN UP' button. At the bottom, there is a link that says 'Already have an account? [Login here](#)'.

Appendix 3: Registration Page

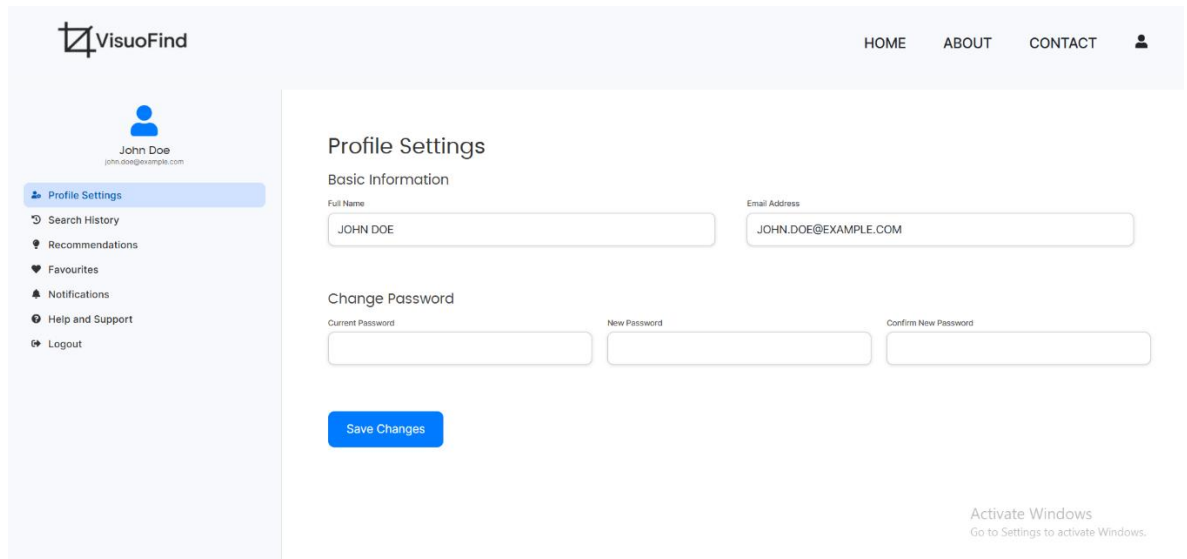


The image shows the 'Search with Image' section. At the top is the title 'Search with Image' with a camera icon. Below it is a large area for uploading an image. It says 'Click or drop image here' with a small icon. Below that is a small text line: 'Exemplo: camiseta e Amarelo-Branco-Homem-Da-Tercer-Do-Cafeteiro-Dos-Homens-Grandes-e-Mais-Pais.jpg'. In the center is a preview of a long-sleeved sweater with a black and white pattern. Below the preview is a blue progress bar and a 'CROP & CONTINUE' button. At the bottom is an 'ADD IMAGE DESCRIPTION (OPTIONAL)' input field and a 'SEARCH' button. On the right side, there is a link that says 'Activate Go to Setting'.

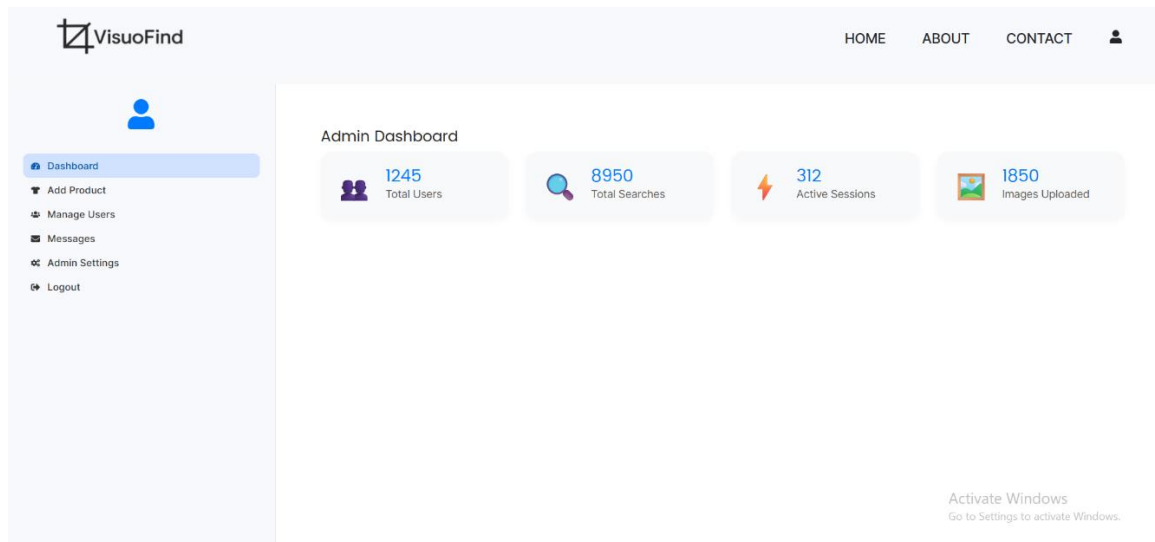
Appendix 4: Image Search Section



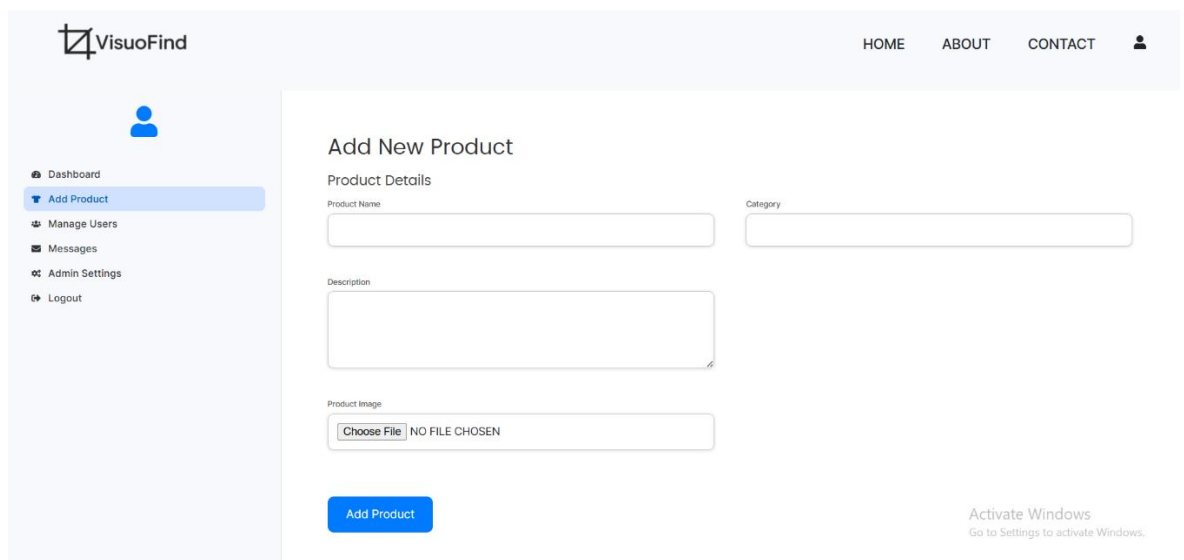
Appendix 5: Recommendation Section



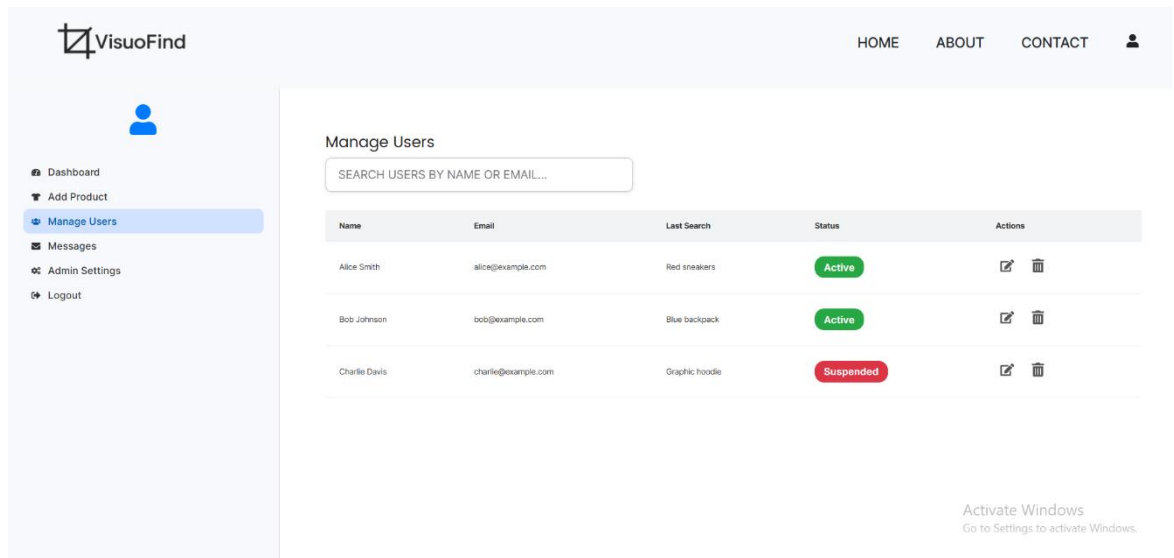
Appendix 6: User Profile Page



Appendix 7: Admin Dash Board



Appendix 8: Add New Product Page



Appendix 9: Manage Users